

Omar Shaker

Backend Software Engineer

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Summary

Software Engineering Intern at Australian Research Centre for Interactive and Virtual Environments (Adelaide University, Australia), building browser-based 3D systems for an education-tech platform. Top-ranked CS student at GIU (GPA 4.0/4.0 (German Scale: 0.7) / A+, Excellence Award Scholarship) specializing in backend development with Python, Django, and REST APIs. Built projects ranging from e-commerce platforms to ML models derived from mathematical first principles. Seeking an internship in Software/Backend Engineering.

Education

German International University (GIU) 2025 – 2029 (Expected)
Bachelor of Computer Science & Informatics Cairo, Egypt

- **GPA:** 4.0/4.0 (German Scale: 0.7 / A+) | **Rank:** 3rd / 500+ | **Excellence Award Scholarship**
- **Coursework:** DSA, Java, Computer Organization, Digital Logic Design, OOP, Theoretical CS, Networks, Linear Algebra

Skills

- **Languages & Frameworks:** Python, Django, Java, C#, SQL (PostgreSQL, Microsoft SQL Server), Assembly (MIPS)
- **Tools & Technologies:** REST APIs, Git, GitHub, AWS, Unity, Jupyter, Eclipse, VS Code
- **Machine Learning:** Linear Regression (Normal Equation, Gradient Descent, Ridge), Model Evaluation (MSE, R^2)
- **CS Fundamentals:** Data Structures & Algorithms (Backtracking, DP, Graphs, Trees), OOP Design, System Design, Linear Algebra

Experience

Australian Research Centre for Interactive and Virtual Environments — Adelaide University Jun 2026 – Present
Software Engineering Intern

- Engineering a **3D model for a 2-storey building in Unity (C#)** structured into modular zones, layers, and metadata so future teams can plug in new interactive learning scenarios without rebuilding the core system
- Translating raw architectural drawings/specs into a structured digital asset under remote mentorship from the academic project lead

AWS Cloud Club — GIU Nov 2025 – Present
Logistics Lead & Core Team Member

- Co-organized **4 sessions, 3 workshops and 2 events at GIU and AUC**, reaching **300+ students**
- Sourced and onboarded **3+ industry experts** for technical talks across online and offline sessions

Technical Content Creator — LinkedIn Dec 2025 – Present
"Python in 60 Seconds"

- Published **60+ posts** of daily Python content simplifying complex concepts, maintaining a daily streak for **3+ months**
- Grew LinkedIn following from **<10 to 230+ (+350%)** over the course of **50 days**

Technical Interview Prep — LeetCode Ongoing

- Solved **110+ problems (Medium/Hard focus)** across trees, graphs, DP, backtracking, and databases

Projects

Store Hub | Python, Django, SQL

- Built a full-stack e-commerce platform across **7 Django apps with 30+ views, 12+ database models**, and a merchant dashboard covering product, order, and inventory management
- Designed a normalized relational schema and implemented RESTful backend logic with authentication, session management, and role-based access across the full user-merchant flow

DooR Dash Board Game | Java, JavaFX, OOP

- Engineered a **2-player, 100-cell** strategy board game across **20+ Java classes with 10+ distinct cell effect types**, leveraging inheritance, polymorphism and abstraction for a fully event-driven JavaFX GUI

Machine Learning: Linear Regression | Python, NumPy, Pandas, Scikit-learn, Jupyter

- Developed Linear Regression from scratch using matrix calculus $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$, achieving machine-precision agreement with scikit-learn (**<1e-13 error**)
- Implemented Gradient Descent with convergence to optimal weights ($\Delta < 1e-3$) and evaluated performance using **MSE and R^2 metrics ($\approx 913 / 0.734$)**
- Extended model with Ridge Regularization, experimentally analyzing bias-variance tradeoff across λ **values (0 to 100)**

Certifications

- Master the Coding Interview: Data Structures + Algorithms | Udemy
- SQL Full Course From Zero to Hero | Data with Baraa
- AWS AI Practitioner Challenge | Udacity (In Progress)
- McKinsey Forward Program | McKinsey Academy (In Progress)